AIORI HACKATHON 3 · SMART CITIES

A3-PS003-TC008

FedTraffic

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

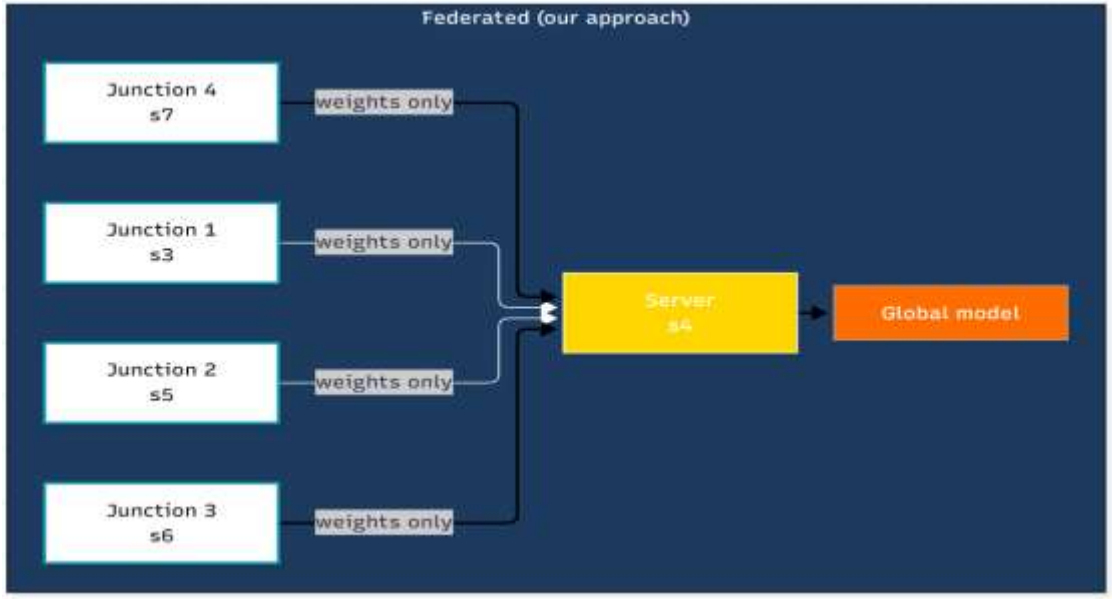
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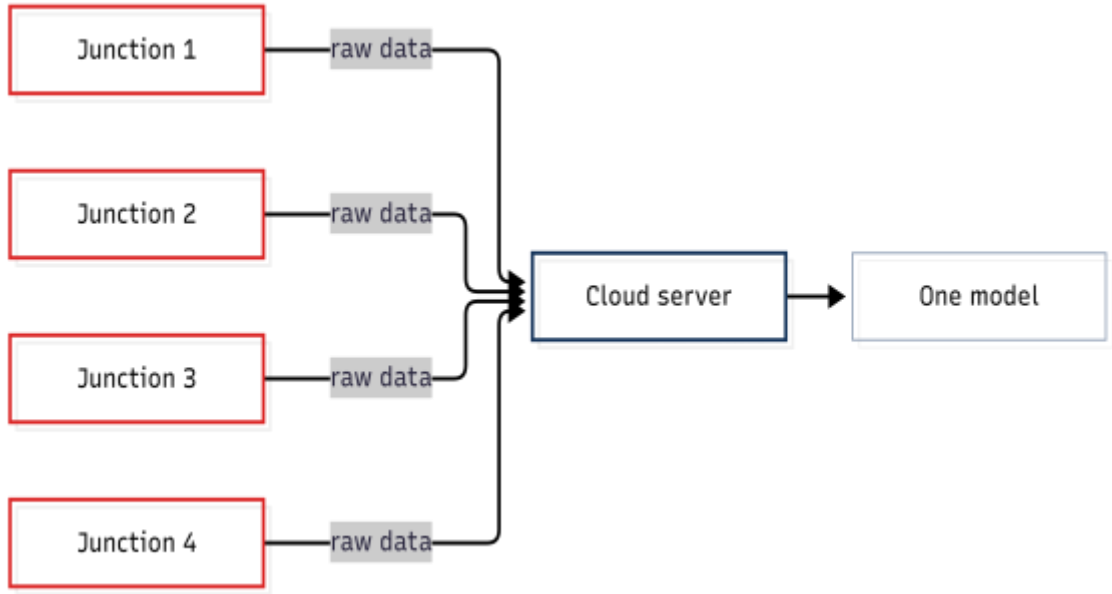
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01 Problem Description

- Traffic forecasting normally streams raw sensor data to a central cloud: heavy on bandwidth, and private travel logs end up in one place.
- Federated learning keeps data at each junction and shares only model weights.
- Our question: how accurate is it when junctions differ (non-IID), and what does each round cost in bytes?
- Scope: a simulation. No live cameras, no city-wide deployment, no formal privacy guarantees.



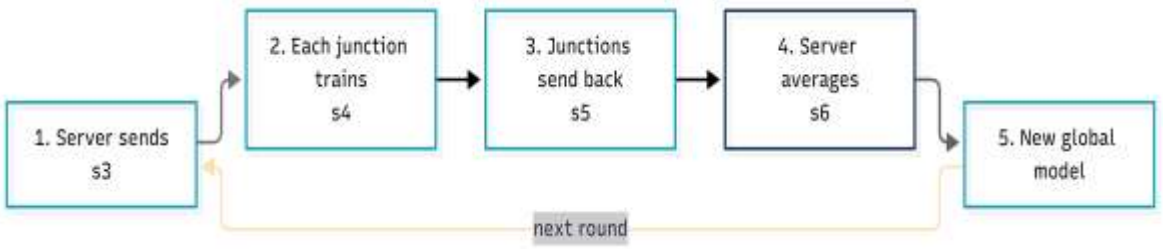
Raw data leaves every junction: heavy on bandwidth, private logs in one place.

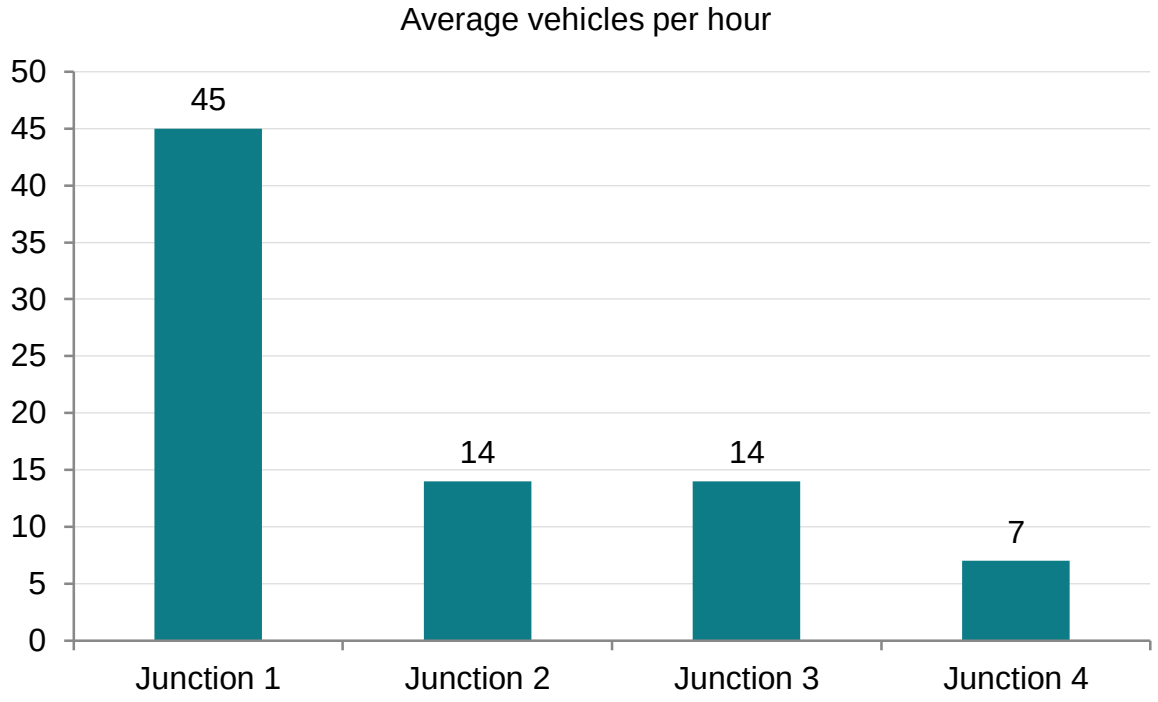


Only model weights leave. Raw data stays at the junction.

02 Background

- **FedAvg** : the server sends the model, clients train locally, and the server averages the returned weights.
- **IID vs non-IID:** non-IID means clients hold different data. Here, junction traffic differs a lot.
- **Why it matters:** federated learning's core challenges include communication cost and statistical heterogeneity.
- **Our metrics:** MAE, RMSE, bytes per round, convergence.





Junction 1 carries about six times the traffic of Junction 4. That's why averaging models across them is hard.

03 Literature & RFC Review

Prior work focuses on accuracy and privacy. We add a controlled study of IID vs non-IID and communication bytes per round on a small, reproducible simulator

1

McMahan et al.

AISTATS 2017

"Communication-Efficient Learning of Deep Networks from Decentralized Data"

- Introduces FedAvg - the aggregation algorithm at the core of our federated loop.

2

Li at al

IEEE SPM 2020

"Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting"

- IEEE survey of federated learning's challenges and approaches
- The paper that introduced the METR-LA dataset we build and evaluate on.
- Frames our focus on communication and non-IID data

3

Sattler et al.

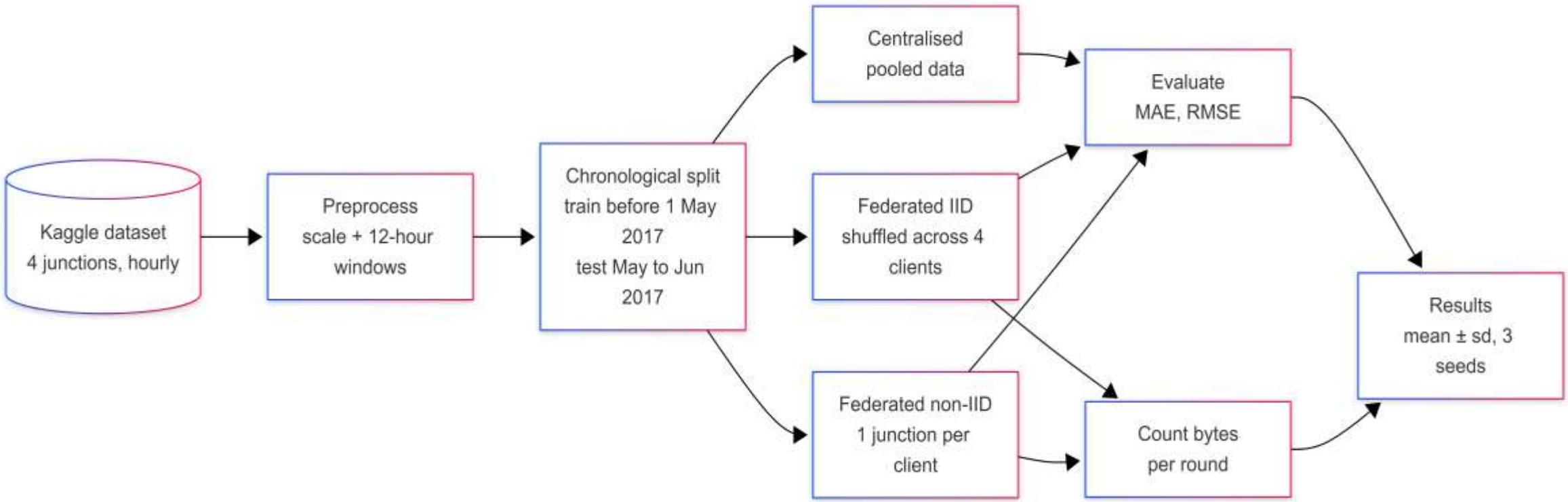
IEEE TNNLS 2020

"Federated Learning with Non-IID Data"

- Notes that federated learning has significant communication overhead, and that non-iid client data is the realistic case where simple methods struggle
- Motivates measuring bytes and testing non-IID

04 Proposed Solution

Same data and same model in all three setups. The only thing that changes is how training happens.

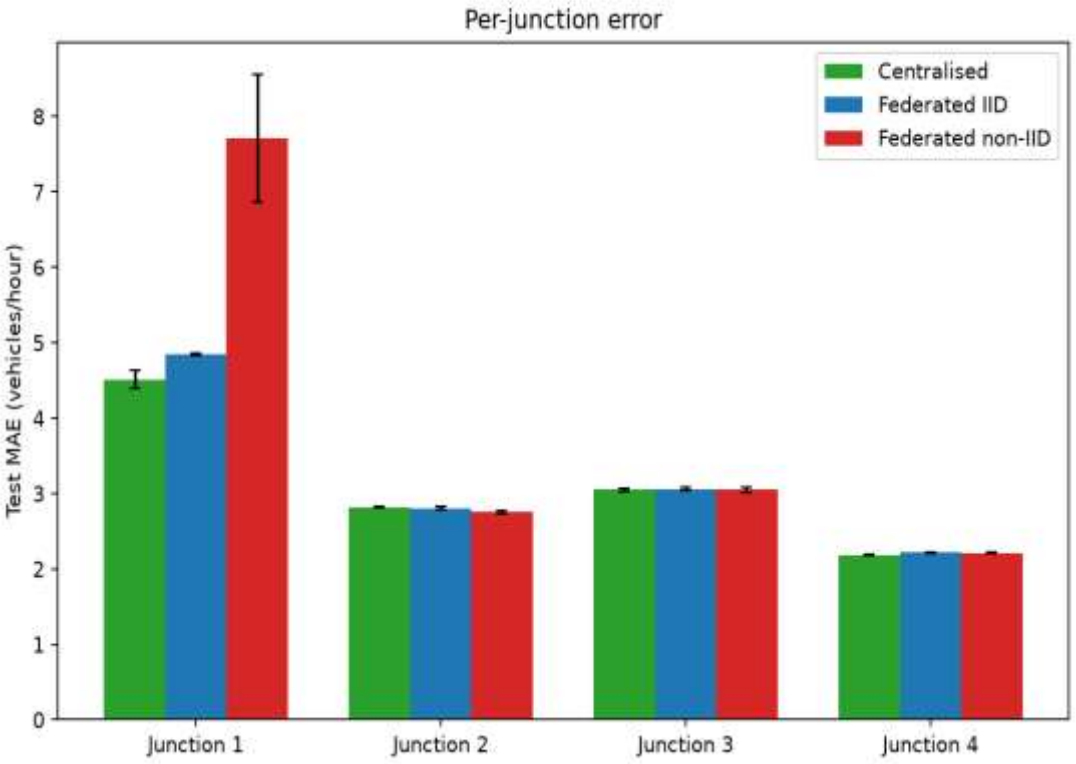


05 Solution Optimization

Take Aways -

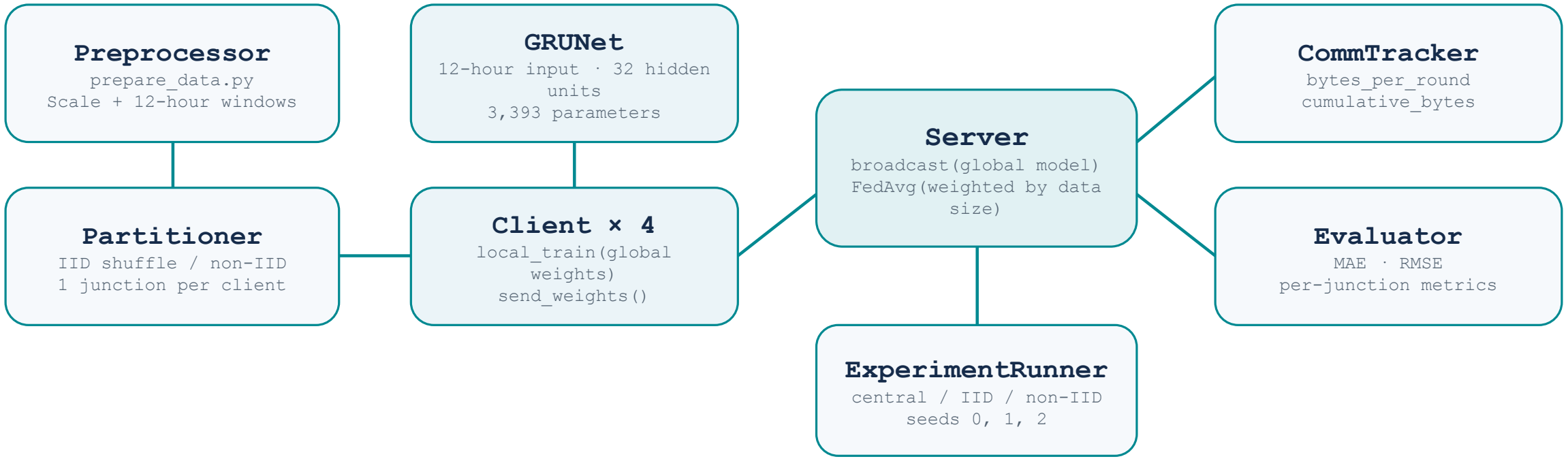
1. IID federated is about 3% worse than centralised.
2. Non-IID is about 25% worse and less stable.
3. The damage is mostly at Junction 1: MAE 4.5 → 7.7.
4. 30 rounds cost about 3.1 MB, more than the roughly 165 KB raw data (approximate). The benefit here is data staying local.

	Central	Fed IID	Fed non-IID
MAE	3.141±0.026	3.232±0.004	3.931±0.201
RMSE	4.927±0.047	5.109±0.012	6.354±0.348
Rounds to converge	~7 epochs	~16	~20
Bytes per round	n/a	108,576	108,576
Raw data leaves	Yes	No	No



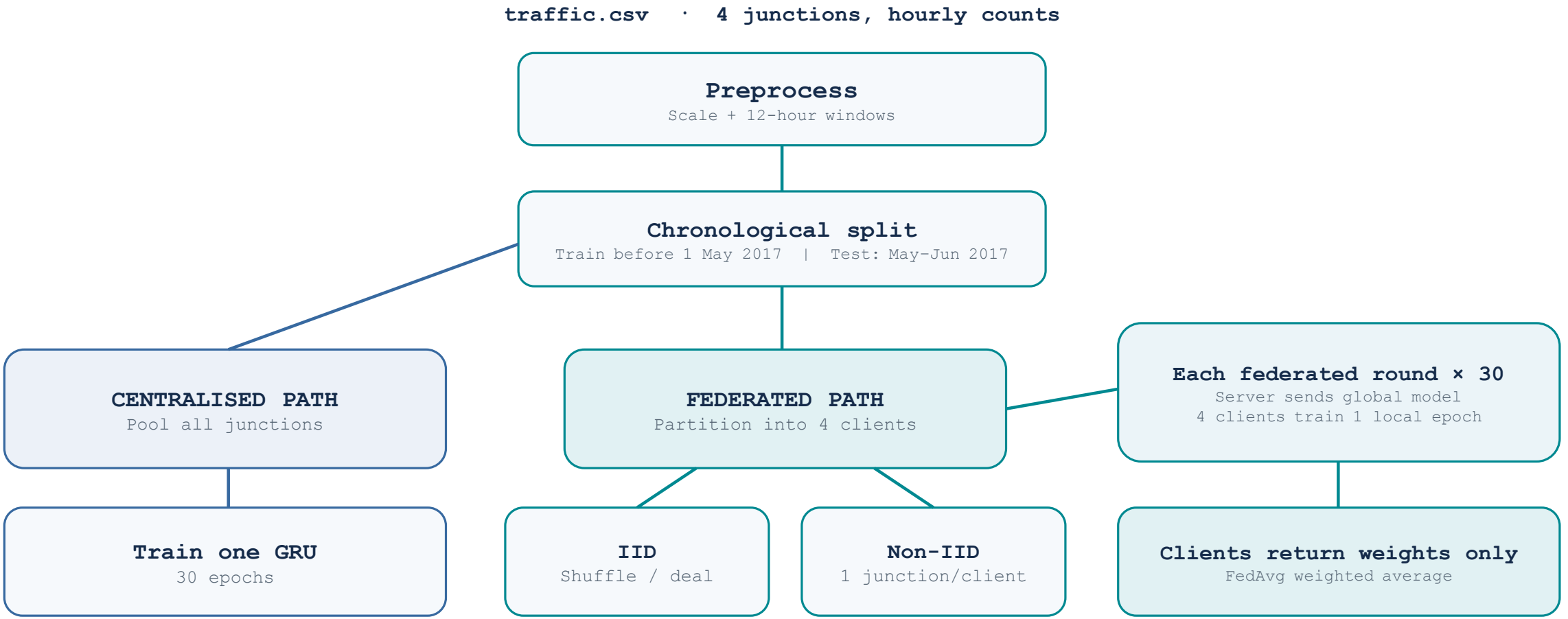
06 Solution Architecture & Design

Logical architecture | Components are logical modules implemented across the project scripts



Data stays at each client. Only model weights/updates are exchanged; evaluation records prediction errors and communication cost.

07 Data Flow Diagram

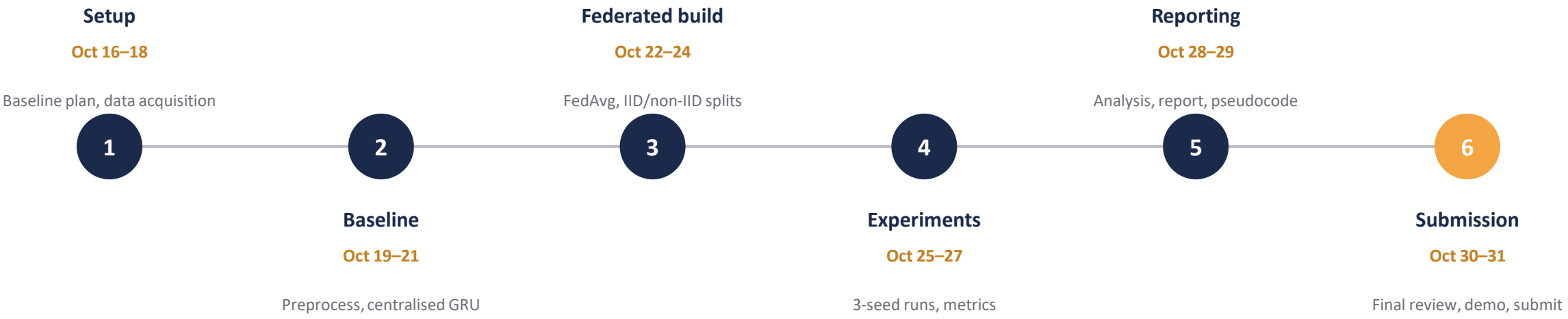


Evaluate on test set: MAE · RMSE · per-junction error | Mean ± SD over seeds 0, 1, 2

Communication tracked: 108,576 bytes per federated round • Raw traffic data never leaves clients

08 Deployment Timeline

Level 2 build window: 16–31 October. As a two-person team, we split ownership so both can work in parallel — one on modelling and the federated loop, the other on data, evaluation and reporting.



Steps 1–7 (GitHub setup, PDF, intro video) are already complete as part of Level 1, due 14 October.

09 Bibliography

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IEEE journal metadata and DOI details verified against publisher/index records; dataset citation included for reproducibility.



Thank You

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